

```
print('Hello')
```

Hello

```
print("It's")
```

It's

```
print('It\'s')
```

It's

```
print('Nashit\nHumam')
```

Nashit

Humam

```
print('Nashit\tHumam')
```

Nashit      Humam

```
print('Nashit\rHum')
```

Humhit

```
print(a)
```

```
-----
-----
NameError                                Traceback (most recent call
last)
Cell In[11], line 1
----> 1 print(a)
```

NameError: name 'a' is not defined

```
for i in range(10):
    print(i)
```

```
Cell In[14], line 2
    print(i)
    ^
```

IndentationError: expected an indented block after 'for' statement on line 1

```
print(12/0)
```

```
-----
-----
ZeroDivisionError                        Traceback (most recent call
last)
Cell In[15], line 1
```

```
----> 1 print(12/0)
```

ZeroDivisionError: division by zero

```
if x=2:
print('hello')
```

```
Cell In[16], line 1
      if x=2:
      ^
```

SyntaxError: invalid syntax. Maybe you meant '==' or ':=' instead of '='?

```
print('hello')
```

```
-----
-----
NameError                                Traceback (most recent call
last)
```

```
Cell In[17], line 1
----> 1 print('hello')
```

NameError: name 'print' is not defined

```
import hello
```

```
-----
-----
ModuleNotFoundError                    Traceback (most recent call
last)
```

```
Cell In[18], line 1
----> 1 import hello
```

ModuleNotFoundError: No module named 'hello'

```
with open('new.txt', 'r') as f:
    d = f.read()
```

```
-----
-----
FileNotFoundError                      Traceback (most recent call
last)
```

```
Cell In[19], line 1
----> 1 with open('new.txt', 'r') as f:
      2     d = f.read()
```

File ~\AppData\Local\Programs\Python\Python312\Lib\site-packages\IPython\core\interactiveshell.py:343, in \_modified\_open(file, \*args, \*\*kwargs)

```
    336 if file in {0, 1, 2}:
    337     raise ValueError(
```

```

338         f"IPython won't let you open fd={file} by default "
339         "as it is likely to crash IPython. If you know what
you are doing, "
340         "you can use builtins' open."
341     )
--> 343 return io_open(file, *args, **kwargs)

```

FileNotFoundError: [Errno 2] No such file or directory: 'new.txt'

```

with open('python.txt', 'r') as f:
    d = f.read()
    print(d)

```

## Module-1: Introduction to Python

Introduction to Python,  
 Installing Python,  
 IDLE, VS Code setup,  
 Structure of a basic Python program,  
 Indentation and Statements in Python,  
 Functions,  
 print() function and escape sequences (\n, \t),  
 main function convention (if \_\_name\_\_ == '\_\_main\_\_'),  
 Calling one function inside another, Order of statements in a  
 function,  
 Comments in Python (#, ''' '''),  
 Variables in Python,  
 Data Types in Python (int, float, bool, str, NoneType),  
 String data type and operations,  
 Type casting (implicit & explicit casting)

## Module-2: User Input, Functions and Control Flow

User Input: input() function,  
 Formatted Output: print() with f-strings, .format(), % formatting,  
 Tokens and string splitting (split(), strip()),  
 Conditional Statements: if, if-else, if-elif-else,  
 Operators in Python (arithmetic, relational, logical, assignment,  
 membership, identity, bitwise),  
 Loops: for, while, for-else, while-else, for with range(), For-each  
 style iteration (lists, strings, tuples),  
 Scope and Lexical Scope (LEGB rule),  
 Introduction to Lists & common list methods,  
 Introduction to Functions in Python,  
 Function arguments: positional, keyword, default, variable-length  
 (\*args, \*\*kwargs),  
 return statement

## Module-3: Data Structures

Garbage Collection and Memory Management in Python,



Cell In[26], line 1

```
----> 1 conn = mysql.connector.connect(
      2     host='localhost',
      3     user='root',
      4     password = 'nashi'
      5 )
```

File ~\AppData\Local\Programs\Python\Python312\Lib\site-packages\mysql\connector\pooling.py:322, in connect(\*args, \*\*kwargs)

```
319     raise ImportError(ERROR_NO_CEXT)
321 if CMySQLConnection and not use_pure:
--> 322     return CMySQLConnection(*args, **kwargs)
323 return MySQLConnection(*args, **kwargs)
```

File ~\AppData\Local\Programs\Python\Python312\Lib\site-packages\mysql\connector\connection\_cext.py:142, in CMySQLConnection.\_\_init\_\_(self, \*\*kwargs)

```
140 if kwargs:
141     try:
--> 142         self.connect(**kwargs)
143     except Exception:
144         self.close()
```

File ~\AppData\Local\Programs\Python\Python312\Lib\site-packages\mysql\connector\abstracts.py:1604, in MySQLConnectionAbstract.connect(self, \*\*kwargs)

```
1601     self.config(**kwargs)
1603 self.disconnect()
-> 1604 self._open_connection()
1606 charset, collation = (
1607     kwargs.pop("charset", None),
1608     kwargs.pop("collation", None),
1609 )
1610 if charset or collation:
```

File ~\AppData\Local\Programs\Python\Python312\Lib\site-packages\mysql\connector\connection\_cext.py:360, in CMySQLConnection.\_open\_connection(self)

```
358 except MySQLInterfaceError as err:
359     if hasattr(err, "errno"):
--> 360         raise get_mysql_exception(
361             err.errno, msg=err.msg, sqlstate=err.sqlstate
362         ) from err
363     raise InterfaceError(str(err)) from err
365 self._do_handshake()
```

ProgrammingError: 1045 (28000): Access denied for user  
'root'@'localhost' (using password: YES)

conn

```
l = [4, 7, 2]
```

```
l[3]
```

```
-----  
-----  
IndexError                                Traceback (most recent call  
last)  
Cell In[28], line 1  
----> 1 l[3]
```

```
IndexError: list index out of range
```

```
a = 78
```

```
b = nashit
```

```
-----  
-----  
NameError                                Traceback (most recent call  
last)  
Cell In[29], line 2  
      1 a = 78  
----> 2 b = nashit
```

```
NameError: name 'nashit' is not defined
```

```
try:  
    print(nashit)
```

```
except:  
    pass
```

```
print(4+6)
```

```
10
```

```
if 2==2:  
    print('rdtr')
```

```
rdtr
```

```
if 2==2:  
    pass
```

```
if 2==8:  
    print('trdtg')
```

```
try:  
    # print(6+23)  
    print(12/0)  
    # print(60+23)
```

```
except:  
    print('division by zero not allowed')
```

division by zero not allowed

```
a = int(input('enter a number'))  
b = int(input('enter second number'))
```

```
try:  
    print(a/b)  
    print(x)  
except:  
    print('zero not allowed')
```

enter a number 6  
enter second number 4

1.5  
zero not allowed

```
a = int(input('enter a number'))  
b = int(input('enter second number'))
```

```
try:  
    print(a/b)  
    print(x)  
except Exception as e:  
    print(e)
```

enter a number 6  
enter second number 0

division by zero

```
try:  
    print(12/0)  
except ZeroDivisionError as e:  
    print(e, 'not allowed')
```

division by zero not allowed

```
try:  
    l = [5, 8, 9]  
    print(l[4])  
except Exception as e:  
    print(e)
```

list index out of range

```
try:  
    s = {'name': 'Nashit'}
```

```
    print(s['age'])
except KeyError as e:
    print(e)
```

```
'age'
```

```
a =int('abc')
```

```
-----
-----
ValueError                                Traceback (most recent call
last)
```

```
Cell In[58], line 1
----> 1 a =int('abc')
```

```
ValueError: invalid literal for int() with base 10: 'abc'
```

```
try:
    a = int('abc')
    print(a)
except Exception as e:
    print(e)
```

```
invalid literal for int() with base 10: 'abc'
```

```
s={'name': 'Nashit'}
print(s['age'])
```

```
-----
-----
KeyError                                Traceback (most recent call
last)
```

```
Cell In[57], line 2
      1 s={'name': 'Nashit'}
----> 2 print(s['age'])
```

```
KeyError: 'age'
```